Donghyung Lee

DevOps Engineer

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Introduction

Hello, I am Donghyung Lee, a DevOps Engineer who designs infrastructure from the ground up and leads its transitions.

Rather than maintaining what was already there, I have mostly led greenfield builds and large-scale migrations. Starting with an OpenStack-based private cloud for financial and public-sector clients, I went on to lead the infrastructure side of a company-wide AWS → GCP migration at a startup, cutting cloud cost 50%. Today I am the sole owner of the AWS + on-premises hybrid infrastructure at a game studio, where I cut monthly infrastructure cost 20% and solely built and launched the EKS production environment for a new game.

My main strength is CI/CD automation, with observability layered on top for production stability. I have built CI/CD pipelines on GitLab CI, GitHub Actions and Jenkins with ArgoCD, and today on Kubernetes I fully automated build and deployment with GitLab CI and ArgoCD GitOps, cutting deployment time 67% (30min → 10min). New components are 100% codified with Terraform and Ansible. On top of that, a 3-signal observability stack connects logs, metrics and distributed traces in Grafana, so an incident can be traced to its cause from a single view.

Beyond that, I build in-house DevOps tools myself to remove operational bottlenecks, and I have worked on AI/ML workload infrastructure as well. At the startup I standardized and scaled an image-generation GPU fleet across 10 in-house boxes, GCP A100 and L4, and 20 external-IDC servers; today I run a docker compose-based in-house LLM inference service on an on-prem GPU server, with a coding model kept loaded in GPU memory and open to the dev team.

I set myself the principle that systems and documents — not one person — should do the work. I believe infrastructure is finished not when it ships but when a fellow developer has a guide they can use on their own, and I keep building environments where solid automation lets the whole team move fast with confidence.

Strengths

Asking 'Why' First
- Ask why the work is needed, define the business problem, then pick the tool

Systematic Documentation & Knowledge Sharing
- Document and share every task, building organizational knowledge assets

Record-Based Continuous Improvement
- Record incidents, infra changes and troubleshooting as postmortems the team reviews together — preventing recurrence, driving improvement

Full-Stack Infrastructure Experience
- End-to-end infra across cloud (AWS, GCP), on-premises, networking, CI/CD, and monitoring

Skills

Cloud	AWS, GCP, OpenStack
Container Orchestration	Kubernetes, Docker, Cilium CNI, NGINX Gateway Fabric / Gateway API, Karpenter
Infrastructure as Code	Terraform, Ansible, Helm, Helmfile
CI/CD	GitHub Actions, GitLab CI/CD, ArgoCD, Argo Rollouts, Jenkins, GitOps
Monitoring & Observability	PLG Stack (Prometheus/Loki/Grafana), ELK Stack, EFK Stack, ECK Operator, Fluent-bit, Thanos, OpenTelemetry, Grafana Tempo
Security & IAM	Keycloak (Operator, OIDC IdP), Vaultwarden, GitLab SSO (OIDC), External Secrets Operator, Sealed Secrets
Automation	Go, Python, Shell Scripting
Database	MySQL, PostgreSQL, Redis, MongoDB
OS & Servers	Ubuntu, Rocky Linux, CentOS, Debian, GPU Server

Professional Experience

Phantom (Concrit Studio)

2024.10 ~ Present

DevOps Engineer

Sole DevOps infrastructure engineer at a game studio, designing, building, and operating AWS-based cloud and on-premises servers. Codified infrastructure with Terraform/Ansible (100% IaC for new components, excluding IAM) and documented operations via automation scripts and postmortems.

- Cut monthly infra cost 20% — sole design and build of the AWS + on-prem hybrid, 100% IaC on new components (excluding IAM)
- Sole-built the launch infrastructure for a new game, expanding a single on-prem dev cluster into an on-prem + AWS multi-cluster (EKS platform 100% GitOps excluding IAM, Karpenter autoscaling). A second cluster on EKS Auto Mode then cut the compute layer's Terraform-managed resources by 83% (102 → 17), and a WireGuard site-to-site VPN gave on-prem a private route
- Built Metrics · Traces · Logs 3-signal observability — log search on one Kibana interface (a 3-stage redesign: ELK → EFK → ECK Operator), OpenTelemetry distributed tracing (OTel Collector · Tempo) and kube-prometheus-stack
- Extended the EFK log pipeline into a product-analytics platform — ES Transform aggregating events per user and Kibana serving D+1~D+30 retention and cohorts, with no commercial analytics SaaS
- Automated manual build-and-deploy with GitLab CI + ArgoCD pull-GitOps, cutting deployment time 67% (30min → 10min); shared toolchain images cut CI provisioning ~97% (5min → ~10s), plus Argo Rollouts Canary and infra automation ported to Python
- Consolidated multiple ingress-nginx instances onto NGINX Gateway Fabric, central SSO for Harbor · ArgoCD · Vaultwarden via Keycloak Operator, plaintext DB passwords removed via External Secrets Operator, static IAM keys removed via IRSA/Pod Identity and static SSH keys via SSM
- Built solo over 6 weeks a self-service portal placing 7 internal apps behind a single OIDC authentication gate, plus 5 operations services (accounts · Keycloak · Harbor · Elasticsearch · VPN) wired to 10 internal account systems, with permissions scoped per team — operational requests that had to pass through one engineer are now handled by each team directly, within its own permissions

Key Achievements

- 20% Monthly Infra Cost Reduction (savings reinvested in new projects)
- Greenfield EKS Sole-Built Production Launch Infra (single on-prem dev cluster → on-prem + AWS multi-cluster, serving real traffic since the 2026.07 soft launch)
- 67% Deployment Time Reduction (30min → 10min, faster market response)

NerdyStar

2023.03 ~ 2024.07

DevOps Engineer

Led the infrastructure side of a large-scale AWS → GCP migration at a game/blockchain startup and built GitOps CI/CD pipelines.

- AWS → GCP large-scale migration (Shared VPC + Workload Identity Federation keyless GitHub Actions auth, subdomain-delegated DNS cutover during a maintenance window, Cloud Armor + Cloud CDN, self-hosted NFS replacing Filestore)
- Built GitOps CI/CD pipeline on GitLab CI · GitHub Actions · Jenkins · ArgoCD
- Built game/blockchain data pipeline (MongoDB · CloudSQL · GA · Dune → BigQuery → Google Sheets, Cloud Functions + Scheduler + Dataflow, Terraform IaC) — replacing analysts' 2-3 hours of daily manual collection with automated loading
- Unified monitoring (PLG Stack: Prometheus · Loki · Grafana) and certificate automation (Cert-Manager + Let's Encrypt)
- Designed & built GPU infrastructure for image-generation AI workloads — 10 in-house boxes, GCP A100 and L4, 20 external-IDC servers

Key Achievements

- 50% Cloud Cost Reduction (Fargate → GKE node pools + MSP discount)
- 2-3h Daily Manual Collection Replaced (analyst hand work → automated loading)

Iorchard

2022.04 ~ 2023.03

Infra Engineer

Designed and built PaaS (Kubernetes + OpenStack + Ceph) private clouds for finance/public-sector clients; led operations training.

- Customized PaaS solution design and construction — delivered to 5 clients (Kubernetes HA + OpenStack + Ceph)
- DP (Delivery Partner) Kubernetes operations training program design and delivery (6 months)
- Wrote the planned shutdown and restart runbook for client power work, handed it to field engineers, and recovered restart failures as second-line support
- Operated and optimized OpenStack and Ceph storage clusters, with regular cluster inspection and troubleshooting
- Automated server provisioning with Ansible, building every client's servers from the same playbooks

Key Achievements

- 5 Cases Client PaaS Solution Deployments (Stable HA-based operation)
- 6 Months DP Field Engineer Kubernetes Training (Monthly · 4 hours a session · about 10 people)

ETech System

2022.01 ~ 2022.04

System Engineer

- Hardware Server and SAN storage construction
- Transitioned to DevOps field for career growth

Opensource Projects

Upstream Contributions (140+ merged, 80+ repos)

nginx/nginx-gateway-fabric · kubernetes-sigs/kueue · elastic/elasticsearch · apache/airflow

Kubernetes CRD

k8s-namespace-sync · helios-lb · network-policy-generator · kube-drift

Helm Charts

nginx-gateway-cr · karpenter-cr · aws-storageclass · buildkit

GitHub Actions

Image Tag Updater · Helm Chart Release Action · Go Kubebuilder Test Action · Helm OCI Push · Major Tag Action

Ansible Galaxy

Ansible K8s IAC Tool · Ansible User Management

DevOps Tools

git-bridge · static-file-server · kube-diff · kube-ctx · helm-upgrade-sync

Claude Code Plugins

sensitive-guard · release-guards · shell-portability

Education

Chungnam National University - Public Administration (Transfer)	2018.03 ~ 2020.08
National Institute for Lifelong Education - Business Administration (Academic Credit Bank)	2016.09 ~ 2018.02

Certifications

Engineer Information Processing	2021.11
AWS Certified Solution Architect - Associate (Expired)	2021.11
Network Administrator Level 2	2021.10
Linux Master Level 2	2021.10
Computer Specialist Level 1	2021.04
TOEIC Speaking Test - 130/Intermediate Mid 3 (Expired)	2021.02

Study

Tech blog (somaz.blog, English) — DevOps · Kubernetes · IaC writing for global readers	2025.01 ~ 현재
Tech blog (somaz.tistory.com, Korean) — ongoing DevOps · Kubernetes · IaC learning and operations notes	2023.04 ~ 현재
T101(Terraform) Study	2023.08 ~ 2023.10
AEWS(EKS) Study	2023.04 ~ 2023.06
PKOS(Kubernetes) Study	2023.01 ~ 2023.02
Cloud Security Training	2022.02 ~ 2022.04
Security Solution Operation Specialist Training Program	2021.07 ~ 2021.12